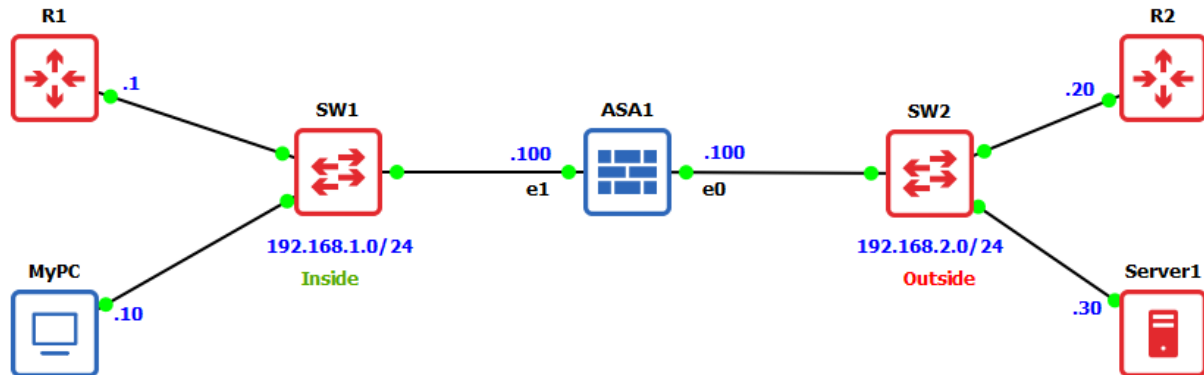


Cisco ASA Basic Configuration:



Cisco ASA Basic Commands	
Commands	Descriptions
ASA>	Unprivileged Mode
ASA#	Privileged Mode
ASA (config)#	Configuration Mode
ASA (config-if)#	Configuration Sub-modes
ASA(config)#configure factory-default	Restores to factory default configuration
ASA# clear configure all	Loads the startup configuration
ASA# write erase	Erases the startup configuration
ASA# show running-config	Views the running configuration
ASA# show startup-config	Views the startup configuration
ASA# copy running-config startup-config	Saves configuration to flash memory
ASA# write mem	Saves configuration to flash memory
ASA(config)# hostname ASA	Changing Host Name
ASA(config)# clear configure hostname	Set default host name
ASA (config)# passwd 123	Set telnet password
ASA (config)# enable password 123	Set enable mode secret Password
ASA (config)# enable password	Remove enable mode password
ASA(config-if)# nameif inside	Assign name to interface
ASA(config-if)#security-level 50	Set interface security level
ASA # show run interface Ethernet 0	Show only interface details
ASA # show nameif	Show interface names
ASA # show interface	Show interfaces
ASA # show interface ip brief	View interfaces briefly
ASA# reload	Reload the ASA
ASA (config)# username admin pass 123 priv 15	Set a username and password
ASA(config)#no service password-recovery	Stop password recovery option
ASA(config)# interface Ethernet 0	Entering interface
ASA(config-if)# ip address 192.168.1.254	Assign IP address to interface
ASA(config-if)# nameif Inside	Given name to interface

Interfaces Setting:

Out of the box, or with the **configure factory-default** command, the ASA 5520 is configured.

Interface	Name	Security Level	IP Address	State
GigabitEthernet0/0 GigabitEthernet0/1 GigabitEthernet0/2 GigabitEthernet0/3	no nameif	no security-level	no ip address	Shutdown
Management0/0	management	100	192.168.1.1	Management-only

- o Interface is named “outside”, ASA automatically assigns the interface security level of 0.
- o If an interface is named “inside”, ASA is automatically assigns a security level of 100.
- o If did not specify subnet mask interface automatically assigned a classful subnet mask.
- o I can specify the subnet mask in the ip address command under the interface configuration.

Interface Configuration
ASA(config)# interface Ethernet1 ASA(config-if)# nameif inside ASA(config-if)# security-level 100 ASA(config-if)# ip address 192.168.1.100 255.255.255.0 ASA(config-if)# no shutdown
ASA(config)# interface Ethernet1 ASA(config-if)# nameif outside ASA(config-if)# security-level 0 ASA(config-if)# ip address 192.168.2.100 255.255.255.0 ASA(config-if)# no shutdown
ASA(config-if)# show running-config interface ASA(config-if)# show running-config interface e0 ASA(config-if)# show running-config interface e1
ASA(config-if)# show ip address
ASA(config-if)# show ip address ethernet 0
ASA(config-if)# show nameif
ASA(config-if)# show nameif ethernet 0
ASA(config-if)# show nameif ethernet 1
ASA(config-if)# show interface ip brief
ASA(config-if)# show interface ethernet 0 ip brief
ASA(config-if)# show interface ethernet 1 ip brief

Device Management:

- o Traffic that network administrator uses to configure network devices is Management.
- o Management plane traffic is usually consists protocol traffic like Telnet, SSH or SNMP.
- o Management plane provides the ability to manage network infrastructure devices.
- o First step toward management is to set and apply the username and the password.

Console Port:

- o Every Cisco Router, Cisco Firewall or a Cisco Switch has a console port.
- o Console port also known as the management port on its backside of device.
- o Console port is used to connect a computer directly to a router or firewall.
- o It manage the router, switch or firewall since there is no display in devices.
- o Console port must be used to initially to install routers, switches & firewalls.
- o There is no network connection initially to connect using SSH, HTTP or HTTPS.
- o Normally Cisco router, Cisco switch or Cisco firewall console port is a RJ45 port.
- o The Console port is the management port, which is used by the administrators.
- o Console port can be used to log into firewall directly without network connection.
- o Console require a terminal emulator application like PuTTY to connect to Firewall.
- o Console port connect to Firewall when a Firewall cannot be accessed over network.

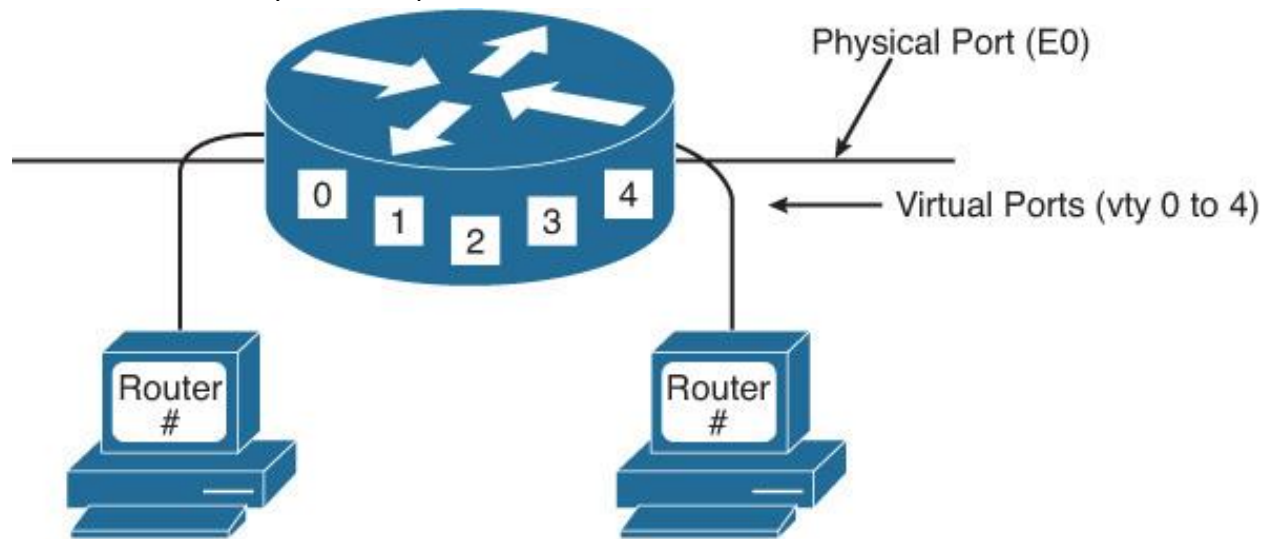


Console Configuration

```
ASA(config)# config t
ASA(config)# username admin password 123 privilege 15
ASA(config)# aaa authentication serial console LOCAL
ASA(config)# exit
ASA# exit
```

VTY (Virtual Terminal Lines):

- o VTY stands for Virtual Terminal Lines or Virtual Teletype.
- o VTY is a Command Line Interface (CLI) created in a router.
- o VTY is used to facilitate a connection to the daemon via Telnet.
- o VTY is just a way of accessing the switches or routers CLI remotely.
- o Virtual Terminals are logical connections from the network to router.
- o VTY are typically telnet connections to switches or routers or Firewall.
- o Telnet is used to manage Cisco routers or switches or firewall remotely.
- o A Telnet client and server application ships with Cisco's IOS software.
- o Telnet commonly uses TCP port 23 to connect to cisco devices router etc.



Telnet Configuration on Cisco ASA

```
ASA(config)# telnet 192.168.1.0 255.255.255.0 inside
```

```
ASA(config)#telnet 192.168.1.1 255.255.255.255 inside
```

```
ASA(config)# telnet 0 0 inside
```

```
ASA(config)#telnet 0.0.0.0 0.0.0.0 inside
```

```
ASA(config)# telnet timeout 10
```

```
ASA(config)# passwd 123
```

```
ASA(config)# username admin password 123
```

```
ASA(config)# aaa authentication telnet console LOCAL
```

```
ASA# who
```

```
ASA(config)# show run telnet
```

```
ASA(config)#clear config telnet
```

SSH (Secure Shell):

- o SSH is a network security term, which stands for Secure Shell.
- o SSH provides secure remote access connection to network devices.
- o SSH (Secure Shell) are two versions SSH Version 1 and SSH Version 2.
- o Communication between client & server is encrypted in both SSH version.
- o SSH (Secure Shell) version 2 is more secure than version SSH Version 1.
- o SSH (Secure Shell) commonly uses TCP port 22 to connect to devices.
- o SSH, as the preferred management protocol under the VTY interfaces.
- o SSH provides a secure and reliable mean of connecting to remote devices.
- o SSH (Secure Shell) Version 2 is more secure and commonly used version.
- o SSH (Secure Shell) require an IOS image that supports crypto features.
- o SSH (Secure Shell) requires RSA public & RSA private key pair to configure.

SSH Configuration on Cisco ASA	
ASA(config)#	crypto key generate rsa modulus 1024
ASA(config)#	ssh 192.168.1.0 255.255.255.0 inside
ASA(config)#	ssh version 2
ASA(config)#	ssh timeout 10
ASA(config)#	passwd 123
ASA(config)#	username admin password 123
ASA(config)#	aaa authentication ssh console LOCAL
ASA(config)#	debug ssh
ASA#	Show SSH sessions
ASA(config)#	show running-config ssh
ASA(config)#	clear config SSH

Verification:

The screenshot displays a network simulator interface with a 'Session Manager' on the left. The main window shows two terminal sessions. The top session is on R1, where the command 'ssh -l admin 192.168.1.100' is entered, followed by the password '123'. The bottom session is on the ASA, where the command 'show ssh sessions' is entered, resulting in a table of active SSH sessions.

SID	Client IP	Version	Mode	Encryption	Hmac	State	Username
0	192.168.1.1	1.99	IN	aes128-cbc	sha1	SessionStarted	admin
			OUT	aes128-cbc	sha1	SessionStarted	admin

HTTP & HTTPS:

- o HTTP is term, which stands for Hyper Text Transfer Protocol.
- o HTTPS is term, which stands for Hypertext Transfer Protocol Secure.
- o Cisco IOS has a HTTP server to managed web-based communication.
- o HTTP Sends the data in clear text over TCP port number 80.
- o HTTP can be used for GUI access on Cisco devices routers & switches.
- o Similar to HTTP, but HTTPs is encrypted over TCP port number 443.
- o Cisco routers permit a user to connect to a router using HTTP.
- o Cisco routers permit a user to connect to a router using HTTPS.

ASDM:

- o ASDM is management tool stand for Cisco's Adaptive Security Device Manager.
- o ASA Security Device Manager (ASDM) is a configuration tool included with ASA.
- o ASA Security Device Manager has easy-to-use Web-based management interface.
- o It is the GUI (Graphical User Interface) tool used to manage Cisco ASA appliances.
- o Offers easy-to-use GUI & enables network administrators to quickly configure ASA.
- o Offers easy-to-use GUI & enables network administrators monitor Cisco ASA Firewall.
- o The ASDM enables network administrators troubleshoot Cisco firewall appliances.
- o ASDM user interface is designed to provide easy access to many features ASA supports.
- o ASDM user interface includes menu bar that provides quick access to files, tools & wizards.
- o A dockable left Navigation pane to move through the Configuration and Monitoring panes.

ASA# copy tftp: disk0:
Address or name of remote host []? 192.168.1.10
Source filename []? asdm-781.bin
Destination filename [asdm-781.bin]?
Accessing tftp://192.168.1.10/asdm-781.bin
ASA# show flash:
ASA# config terminal
ASA(config)# asdm image flash:/asdm-781.bin

HTTPS/ASDM Configuration on Cisco ASA
ASA(config)# username admin password 123
ASA(Config)# http server enable
ASA(config)# http 192.168.1.0 255.255.255.0 inside
ASA(config)# http 192.168.1.1 255.255.255.255 inside
ASA(Config)# http 0.0.0.0 0.0.0.0 inside
ASA(config)# aaa authentication http console LOCAL
ASA# show run http
ASA(config)#clear config http

[illegible]

```
ASA(config)# dir disk0:

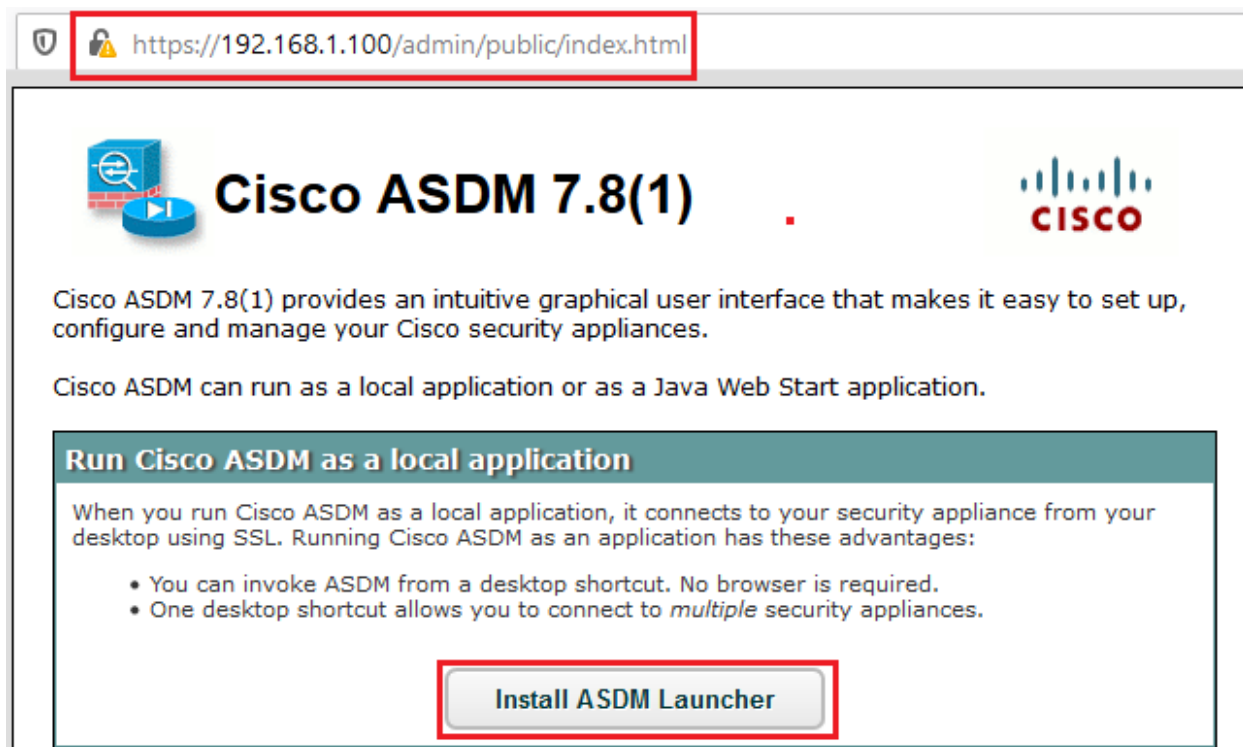
Directory of disk0:/

10      drwx  4096          10:49:58 Mar 01 2013    log
21      drwx  4096          10:50:00 Mar 01 2013    coredumpinfo
99      -rwx  25154848      12:18:24 Feb 05 2018    anyconnect-ma
100     -rwx  26916068      15:09:36 May 05 2020    asdm-781.bin
101     -rwx  28672          00:00:00 Jan 01 1980    FSCK0000.REC
102     -rwx  4096          00:00:00 Jan 01 1980    FSCK0001.REC
103     -rwx  35327062      12:18:46 Feb 05 2018    anyconnect-wi
104     -rwx  20137698      12:18:54 Feb 05 2018    anyconnect-li
105     -rwx   3            01:27:08 May 06 2020    abc.txt
108     -rwx  26916068      08:44:00 May 06 2020    asdm781

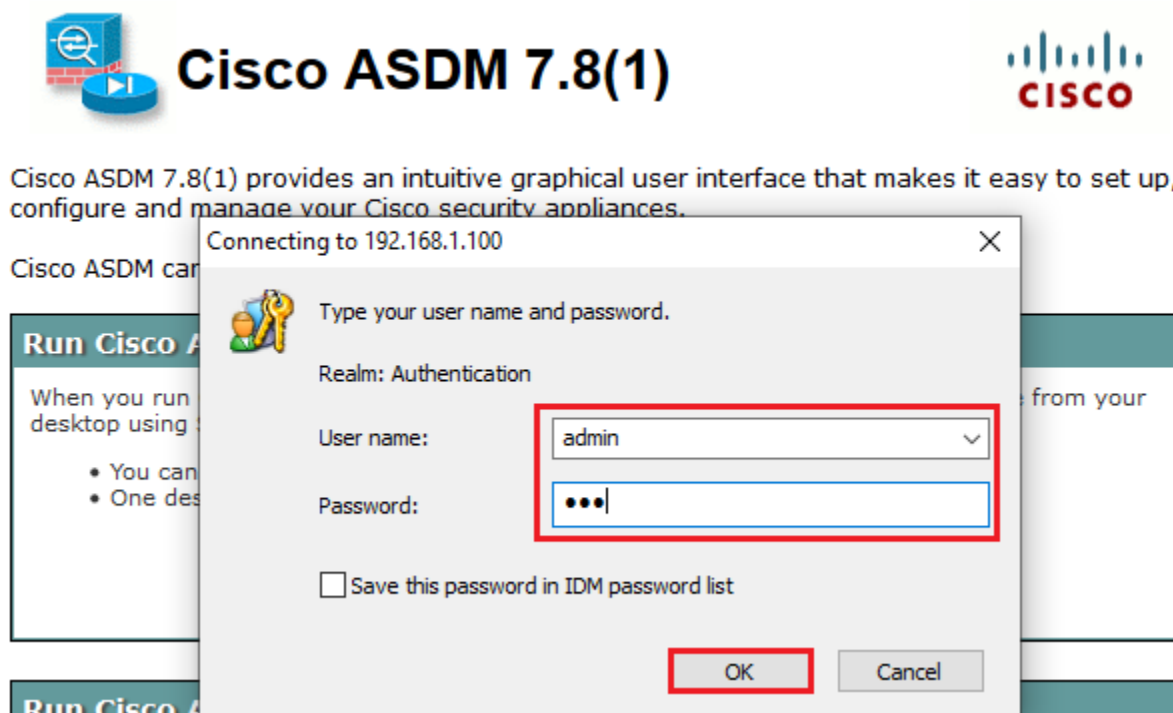
268054528 bytes total (106065920 bytes free)
ASA(config)# asdm image disk0:/asdm781
```

```
ASA(config)# username admin password 123
ASA(config)# http server enable
ASA(config)# http 0 0 inside
ASA(config)# aaa authentication http console LOCAL
ASA(config)# show run http
http server enable
http 0.0.0.0 0.0.0.0 inside
ASA(config)#
```

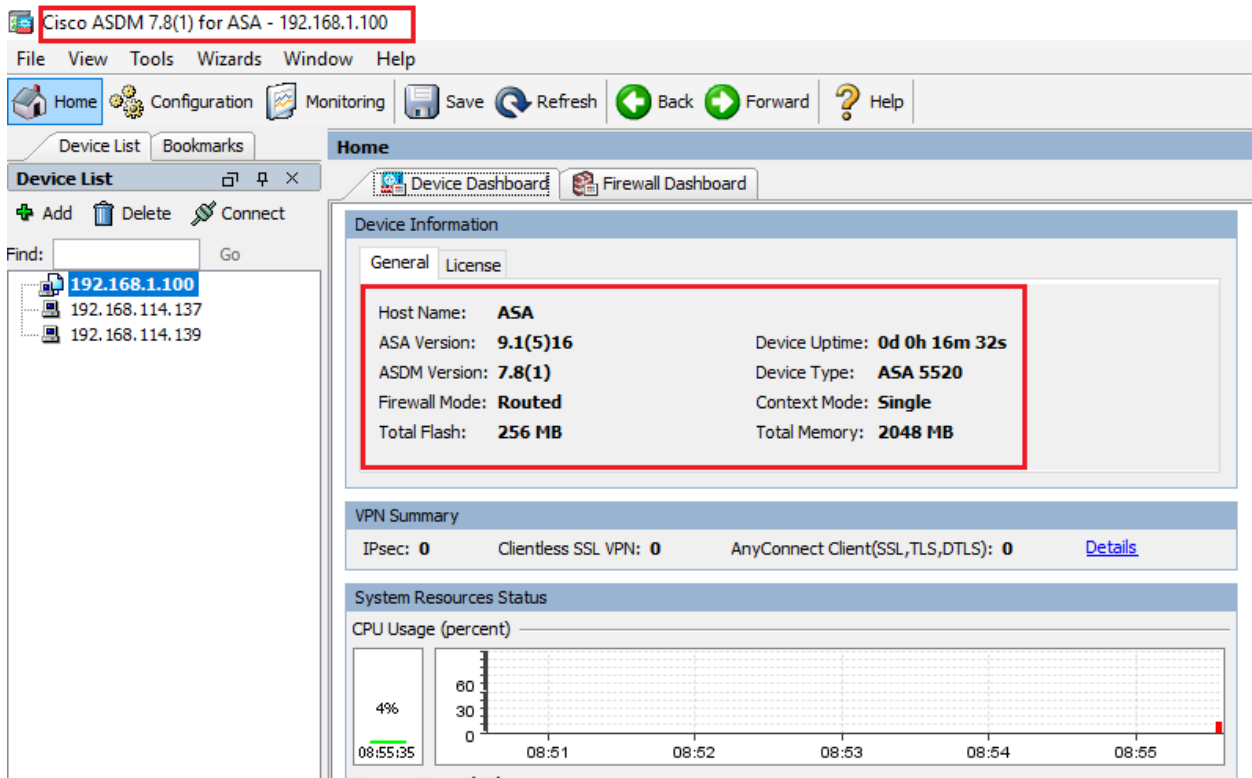
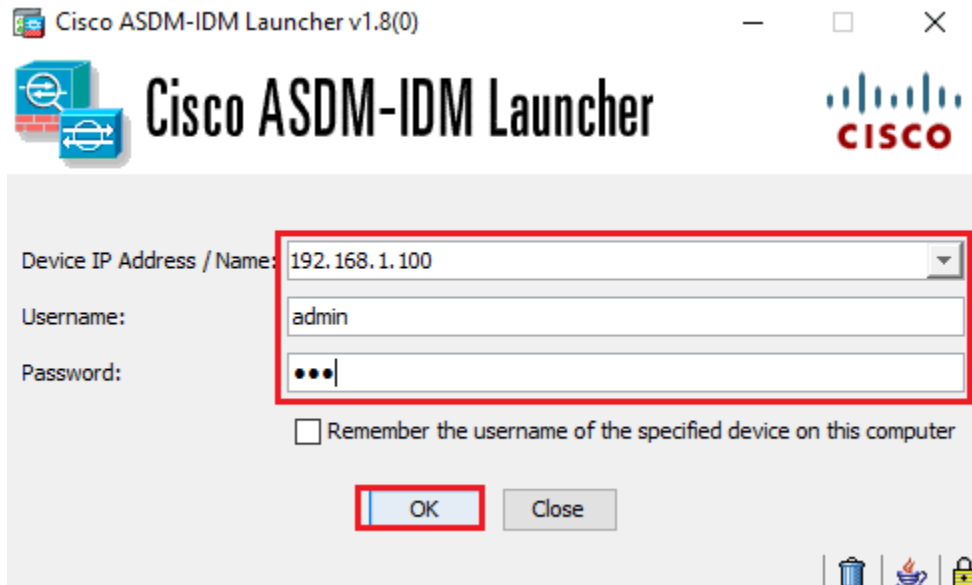

Open browser and go to the IP address of the ASA device use the HTTPS protocol: Click on the Install ASDM Launcher and Run ASDM button to install ASDM.



If prompted for credentials, enter the username and password you've configured.



Run the downloaded installer and finish the wizard. When the installation is completed, you can start using ASDM to manage your ASA:



TFTP (Trivial File Transfer Protocol):

- o TFTP is term, which stand for Trivial File Transfer Protocol.
- o TFTP is used to save & restore a router or firewall configuration.
- o TFTP is used to backup, restore and upgrade IOS image of devices.
- o Trivial File Transfer Protocol uses UDP (User Datagram Protocol).
- o Trivial File Transfer Protocol uses a well-known UDP port no 69.
- o Trivial File Transfer Protocol requires very small amount of memory.
- o TFTP uses connection-less User Datagram protocol for communication.
- o Trivial File Transfer Protocol (TFTP) is a simple file transfer protocol.
- o TFTP does not support user authentication & all data is sent in clear text.
- o Trivial File Transfer Protocol uses less resource as compare to FTP protocol.

FTP (File Transfer Protocol):

- o FTP is a network term, which is stands for File Transfer Protocol.
- o File Transfer Protocol is used to send & receive file from the remote device.
- o File Transfer Protocol is a complete, session-oriented file transfer protocol.
- o FTP depends on TCP, is connection oriented & provides reliable control.
- o FTP (File Transfer Protocol) provides user authentication method.
- o FTP (File Transfer Protocol) uses usernames and passwords for setup.
- o Devices are required to have a username and password setup for FTP.
- o FTP (File Transfer Protocol) is faster when compared to TFTP protocol.
- o It uses two TCP ports 20 for sending data & 21 for sending control commands.

TFTP Commands
ASA(config)# copy running-config tftp:
ASA(config)# copy startup-config tftp:
ASA(config)# copy tftp: running-config
ASA(config)# copy tftp: startup-config
ASA(config)# copy tftp: flash:
ASA(config)# copy tftp: disk0:
ASA(config)# copy disk0: tftp:

FTP Commands
ASA(config)# copy ftp://admin:123456@192.168.1.10/test.txt disk0:/test.txt
ASA(config)# copy disk0:/test.txt ftp://admin:123456@192.168.1.10/mytest.txt